

SIM1001 Basic Mathematics
SIX1013 Fundamental to Advanced Mathematics
Assignment (20%)

Institute of Mathematical Sciences
University of Malaya

Semester 1 2025/2026

Instruction:

- (i) Answer **ALL** questions.
- (ii) Write your NAME and STUDENT ID Number on the FIRST PAGE of your answer scripts. Upload the file for your answer scripts to SPeCTRUM and the file name is SIM1001 CA XXXXXXXXX or SIX1013 CA XXXXXXXXX where XXXXXXXX is your student ID Number (without /1 or /2).
- (iii) Please upload your answer scripts to UM SPeCTRUM by **10am, 11th December 2025 (Thursday)**.

1. Consider the following statement S :

S : For all integer x , x is a prime number and x divides 15525 only if x divides 20250.

- a. Write the statement S and its negation in symbolic form.
- b. What is the truth value of the statement S ? Justify your answer.

2. Let p , q , and r be statements variables. Use the laws of logical equivalences, show that the following of compound statement can be simplified to a simple statement.

$$p \vee \sim \{ [(p \vee q) \wedge r] \rightarrow \sim q \} \vee r \}$$

Give the reasons for each step of simplifications of compound statements.

3. Use **truth table** to determine the validity of the following argument:

This polygon is a not triangle unless the sum of its interior angles is 180° .

A necessary condition for this polygon to be a rectangle is that it is neither a triangle nor a square.

The sum of the interior angles of this polygon is 180° or this polygon is a rectangle, but it is not a square.

$\therefore$ This polygon is triangle.

4. Define the symmetric difference $A \oplus B$ of two sets A and B to be

$$A \oplus B = (A - B) \cup (B - A).$$

- a. Draw a Venn diagram representing $A \oplus B$.
 - b. Let $A = \{ \text{odd number from 1 to 20} \}$ and $B = \{ \text{multiples of 3 from 1 to 20} \}$. Find $A \oplus B$.
 - c. Suppose that A , B , and C are sets such that $A \oplus C = B \oplus C$. Prove that $A = B$.
5. Consider the two functions $f: S \rightarrow T$ and $g: T \rightarrow U$ for non-empty set S, T, U . Determine if the following statement is true or false. Justify your answer.
- a. If $g \circ f$ is one-to-one, then f is one-to-one.
 - b. If $g \circ f$ is one-to-one, then f is onto.
 - c. If $g \circ f$ is one-to-one and f is onto, then g is one-to-one.